

Himanshu Kumar Sah

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Creative Portfolio

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Certified RHCSA candidate with hands-on experience in crypto wallet scrapers, Raspberry Pi Wi-Fi auditing, and a homelab for real-time attack-defense simulation. Creative professional with a club management and mass event background, including managing Front of House at TEDxVITAP 2024 and visual design for 500+ guests. Awarded with podium rankings at national and college-level hackathons.

Experience

MagicBricks

Noida, Delhi (On Site)

Technology Department Intern

Jan 2026 - May 2026

- Researched and integrated Large Language Models (LLMs) to enhance automated workflow processes.
- Engineered system prompts and context-handling strategies to optimize LLM accuracy and reduce response latency.
- Built an enterprise-grade AI outbound calling platform utilizing FastAPI and the LiveKit Agents SDK to automate customer communications.

Netts Mobilitiy

Amarawati, Andhra Pradesh (On Site)

Graphic Designer Intern

Apr 2025 – July 2025

- Create visually appealing designs for NETTS Mobility's marketing materials, website and social media.
- Collaborate with the marketing and product teams to align designs with brand goals.
- Conduct design research to ensure visuals resonate with target audiences.
- Develop graphics, banners, posters, and other assets to enhance the brand's presence.

Projects

Autonomous AI Voice Agent & Telephony Orchestrator

- Architected a high-concurrency outbound calling engine using FastAPI, Celery, and Redis, implementing a PostgreSQL transactional locking strategy (SELECT ... FOR UPDATE SKIP LOCKED in dispatcher.py) to ensure thread-safe, concurrent lead claiming across distributed workers.
- Engineered a low-latency voice AI orchestration pipeline using the LiveKit Agents SDK (agent.py), integrating dynamic STT/TTS models and a pgvector RAG search (using Gemini embeddings) to inject context-aware knowledge base records directly into active audio streams.

Custom Firewall & SIEM Security Platform (Capstone Project)

- Designed and deployed a virtualized enterprise network security topology on Proxmox VE, routing traffic through an OPNsense firewall integrated with Wazuh SIEM to correlate event logs and generate real-time threat alerts.
- Developed a custom React/Node.js telemetry dashboard to monitor network statistics, implementing automated Quality of Service (QoS) bandwidth-shaping rules to dynamically allocate resources and prioritize critical services during peak loads.

Peer-to-Peer Cryptographic File Sharing Application

- Developed a cross-platform peer-to-peer file transfer utility in Python, utilizing raw TCP socket programming to manage low-level network data buffers and optimize transfer throughput over local networks.
- Implemented a secure transmission mode using public-key cryptography to encrypt payload data in transit, ensuring cryptographic integrity and protecting transferred files against local network sniffing.

MERN Stack Food Ordering Web Application (SnackMate)

- Architected a full-stack e-commerce web application using React, Node.js, Express, and MongoDB, implementing secure user authentication and global state management via Zustand to coordinate data flow.
- Designed a separate administrative dashboard and inventory management system, developing RESTful API endpoints with optimized MongoDB query schemas to support real-time CRUD operations and order tracking.

Leadership & Activities

Open Source Community, VIT-AP

Amarawati, Andhra Pradesh

Track Lead

May 2024 – May 2025

- Coordinated cross-functional workflows and task timelines across multiple technical departments, ensuring the smooth execution of club operations, workshops, and open-source contributions.
- Monitored deliverable progress for club members, proactively identifying and resolving development blockers to maintain consistent project momentum.

TEDxVITAP

Amarawati, Andhra Pradesh (On Site)

Head of Front of House (FOH)

April 2024

- Orchestrated end-to-end event logistics, crowd control, and hospitality operations for a high-profile campus event with over 500+ attendees.
- Led and supervised a team of FOH volunteers, delegating responsibilities and managing real-time event operations to ensure a seamless visitor experience.

Education

Vellore Institute of Technology, AP

Amarawati, Andhra Pradesh

B.Tech CSE, Specialization in Cyber Security, GPA 8.13

2022 - 2026

DAV, Ashok Vihar

Delhi, Delhi

CBSE 12th Board Percentage (best of 5): 76.5%

July 2022

Skills & Interests

Languages & Frameworks: Python, HTML, CSS, JavaScript, React.js, Java, Linux

Tools/Platforms: Nmap, Hashcat, Wireshark, Git, Splunk, Wazuh, OPNSense Firewall, Ethical Hacking, Web Application Security

Creative Tools: Figma, Adobe Premiere Pro, Adobe After Effects, Davinci Resolve, Fusion 360, Blender-3D

Languages: English, Hindi, Spanish (Ongoing learning)

Practical Labs: Setting up Windows Server, AD DS & AD CS, adding users, connecting domains, imitating real world Cyber Attacks and their mitigations

Certifications

- RHCSA - RedHat System Administrator
- ISC2 - CC

Courses

- JNCAA - Design, Associate Course
- AWS Academy Cloud Foundations from AWS Academy
- Harvard CS50
- Introduction To Cyber Security - Great Learning Academy
- Cybersecurity Career Starter Certification - Hack & Fix Academy
- WireShark Course by David Bombal - Udemy (Ongoing)